Comprendre l'évolution du stack en Java

Semaine 4

École des Sciences Criminelles - UNIL

Terminologie

On utilise:

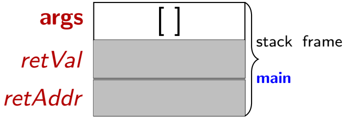
- *retAddr* pour indiquer l'adresse de retour d'une méthode (return address).
- *retVal* pour indiquer la valeur de retour d'une méthode (return value).

On écrit de cette façon afin de rendre les choses plus explicites. Cependant, quand vous dessinez un diagramme du stack, il faut utiliser les conventions dans les diapositives du cours, c'est à dire:

- $\hookleftarrow$ pour l'adresse de retour.
- $\uparrow$ pour la valeur de retour.

`main` — Aucun argument de la ligne de commande

```java
public class LeapYear {
    static boolean isLeap(int year) {
        return (year % 4 == 0) && (year % 100 != 0) || (year %
              400 == 0);
    }
    static void reportLeapYear(int year) {
        boolean leap = isLeap(year);
        System.out.println("isLeap(" + year + ")=" + leap);
    }
    public static void main(String[] args) {
        int year = 2000;
        reportLeapYear(year);
    }
}
```

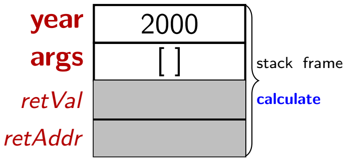


On commence toujours l'exécution dans la méthode `main`!

L'adresse de retour de la méthode `main` est dans le JVM ☕

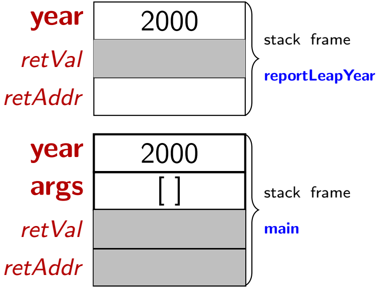
main:L10

```java
1  public class LeapYear {
2      static boolean isLeap(int year) {
3          return (year % 4 == 0) && (year % 100 != 0) || (year %
               400 == 0);
4      }
5      static void reportLeapYear(int year) {
6          boolean leap = isLeap(year);
7          System.out.println("isLeap(" + year + ")=" + leap);
8      }
9      public static void main(String[] args) {
10         int year = 2000;
11         reportLeapYear(year);
12     }
13 }
```



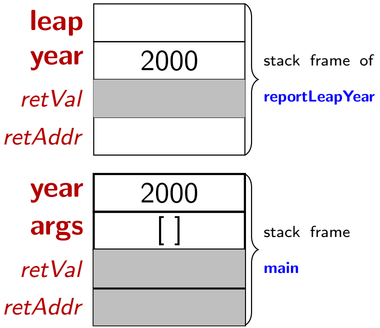
main:L11

```java
1  public class LeapYear {
2      static boolean isLeap(int year) {
3          return (year % 4 == 0) && (year % 100 != 0) || (year %
               400 == 0);
4      }
5      static void reportLeapYear(int year) {
6          boolean leap = isLeap(year);
7          System.out.println("isLeap(" + year + ")=" + leap);
8      }
9      public static void main(String[] args) {
10         int year = 2000;
11         reportLeapYear(year);
12     }
13 }
```



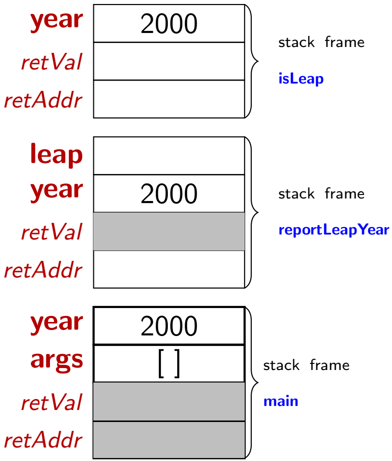
`reportLeapYear`:L6 — Allocation de la variable `leap`

```java
1  public class LeapYear {
2      static boolean isLeap(int year) {
3          return (year % 4 == 0) && (year % 100 != 0) || (year %
               400 == 0);
4      }
5      static void reportLeapYear(int year) {
6          boolean leap = isLeap(year);
7          System.out.println("isLeap(" + year + ")=" + leap);
8      }
9      public static void main(String[] args) {
10         int year = 2000;
11         reportLeapYear(year);
12     }
13 }
```



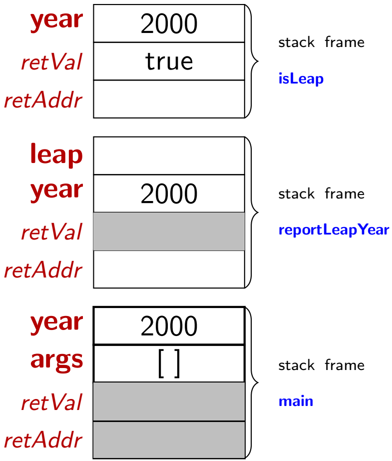
reportLeapYear:L6 — Appel de la méthode isLeap

```
1  public class LeapYear {
2      static boolean isLeap(int year) {
3          return (year % 4 == 0) && (year % 100 != 0) || (year %
               400 == 0);
4      }
5      static void reportLeapYear(int year) {
6          boolean leap = isLeap(year);
7          System.out.println("isLeap(" + year + ")=" + leap);
8      }
9      public static void main(String[] args) {
10         int year = 2000;
11         reportLeapYear(year);
12     }
13 }
```



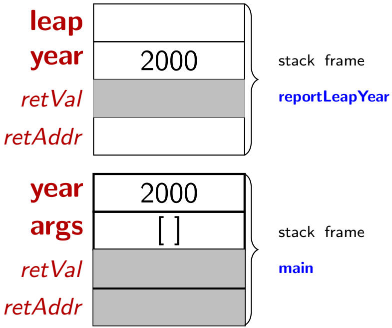
isLeap:L3 — Calcul d'une valeur booléenne

```
1  public class LeapYear {
2      static boolean isLeap(int year) {
3          return (year % 4 == 0) && (year % 100 != 0) || (year %
               400 == 0);
4      }
5      static void reportLeapYear(int year) {
6          boolean leap = isLeap(year);
7          System.out.println("isLeap(" + year + ")=" + leap);
8      }
9      public static void main(String[] args) {
10         int year = 2000;
11         reportLeapYear(year);
12     }
13 }
```



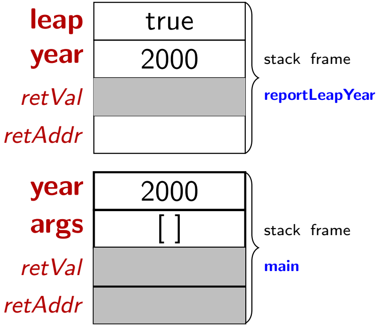
isLeap:L3 — Sortie de la méthode isLeap

```
1  public class LeapYear {
2      static boolean isLeap(int year) {
3          return (year % 4 == 0) && (year % 100 != 0) || (year %
               400 == 0);
4      }
5      static void reportLeapYear(int year) {
6          boolean leap = isLeap(year);
7          System.out.println("isLeap(" + year + ")=" + leap);
8      }
9      public static void main(String[] args) {
10         int year = 2000;
11         reportLeapYear(year);
12     }
13 }
```



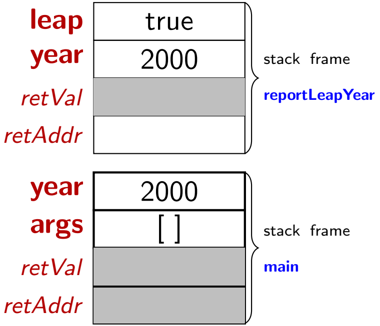
`reportLeapYear:L6` — Affectation de la variable `leap`

```java
1  public class LeapYear {
2      static boolean isLeap(int year) {
3          return (year % 4 == 0) && (year % 100 != 0) || (year %
               400 == 0);
4      }
5      static void reportLeapYear(int year) {
6          boolean leap = isLeap(year);
7          System.out.println("isLeap(" + year + ")=" + leap);
8      }
9      public static void main(String[] args) {
10         int year = 2000;
11         reportLeapYear(year);
12     }
13 }
```



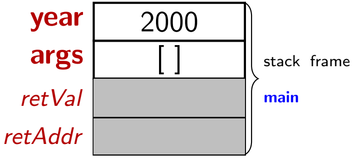
`reportLeapYear`:L7 — Affichage d'un message avec `System.out.println`

```
1  public class LeapYear {
2      static boolean isLeap(int year) {
3          return (year % 4 == 0) && (year % 100 != 0) || (year %
               400 == 0);
4      }
5      static void reportLeapYear(int year) {
6          boolean leap = isLeap(year);
7          System.out.println("isLeap(" + year + ")=" + leap);
8      }
9      public static void main(String[] args) {
10         int year = 2000;
11         reportLeapYear(year);
12     }
13 }
```



`reportLeapYear`:L7 — Sortie de la méthode `reportLeapYear`

```java
1  public class LeapYear {
2      static boolean isLeap(int year) {
3          return (year % 4 == 0) && (year % 100 != 0) || (year %
               400 == 0);
4      }
5      static void reportLeapYear(int year) {
6          boolean leap = isLeap(year);
7          System.out.println("isLeap(" + year + ")=" + leap);
8      }
9      public static void main(String[] args) {
10         int year = 2000;
11         reportLeapYear(year);
12     }
13 }
```



`main`:L11 — Sortie de la méthode `main`

```
1  public class LeapYear {
2      static boolean isLeap(int year) {
3          return (year % 4 == 0) && (year % 100 != 0) || (year %
               400 == 0);
4      }
5      static void reportLeapYear(int year) {
6          boolean leap = isLeap(year);
7          System.out.println("isLeap(" + year + ")=" + leap);
8      }
9      public static void main(String[] args) {
10         int year = 2000;
11         reportLeapYear(year);
12     }
13 }
```

Besoin de plus de visualisations?

Visitez ce site-web génial pour plus de visualisations:

`https://cscircles.cemc.uwaterloo.ca/java_visualize/`